

In addition to the data flows captured above, this specification defines secure session establishment protocols based on operational certificates (see [Section 4.14.2, “Certificate Authenticated Session Establishment \(CASE\)”](#)), or passcodes (see [Section 4.14.1, “Passcode-Authenticated Session Establishment \(PASE\)”](#)), group communication (see [Section 4.16, “Group Communication”](#)), a bulk data transfer protocol (BDX) suitable for sending bulk data between Nodes, and provisions for defining manufacturer-specific protocols.

2.3. Network Topology

In principle, any IPv6-bearing network is suitable for Matter deployment, subject to supporting a few core IPv6 standards. In this version of the specification, we focus on three link layer technologies: Ethernet, Wi-Fi and Thread. We restrict the specification to the above so that the specification can suitably cover provisioning of these link layers, and so that the amount of testing in certification is suitably bounded.

Matter treats networks as shared resources: it makes no stipulation of exclusive network ownership or access. As a result, it is possible to overlay multiple Matter networks over the same set of constituent IP networks.

This protocol may operate in the absence of globally routable IPv6 infrastructure. This requirement enables operation in a network disconnected or firewalled from the global Internet. It also enables deployment in situations where the Internet Service Provider either does not support IPv6 on consumer premises or where the support proves otherwise limiting, for example, if the delegated prefix cannot accommodate all the networks and devices on premises.

This protocol supports local communications spanning one or more IPv6 subnets. Canonical networks supporting a fabric may include a Wi-Fi/Ethernet subnet, or one or more low power and lossy networks (LLN) subnets. In this version of the specification, Thread is the supported LLN standard.

2.3.1. Single network

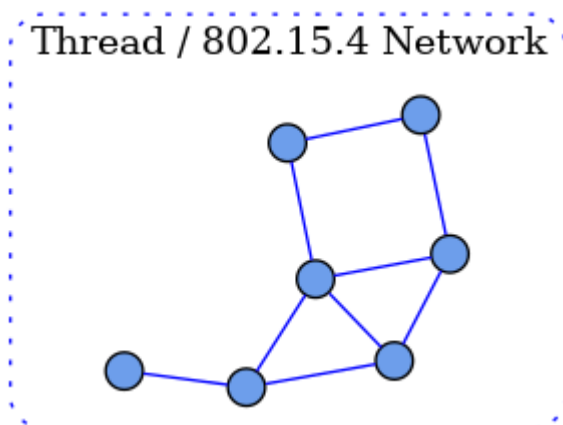


Figure 3. Single Thread network